

Nature Technical Report: Optimization of Deep Brain Stimulation for Dementia and Smart Aging Paradigms

This report documents the architectural specifications and mathematical manifolds governing the advanced Deep Brain Stimulation (DBS) framework for dementia and smart aging paradigms.

1. Comprehensive Electrical Specifications

- Voltage Range: 0.0 V - 10.5 V
- Current Range: 0.0 mA - 25.5 mA
- Rate (Frequency): 2 Hz - 250 Hz
- Pulse Width: 60 μ s - 450 μ s
- Interfacial Range: 500 Ω - 1500 Ω
- Telemetry Bands: MICS (402-405 MHz), ISM (2.4 GHz), mmWave
- Power: Inductive Link ($\sim$ 125 kHz) or Resonant ($\sim$ 6.78 MHz)
- Constraint: Sub-mW limitations strict tissue thermal protection.

2. Cortical Manifolds & Finite Element Modeling

The cortical Volume of Tissue Activated (VTA) is modeled as a Riemannian manifold. We deploy the finite math mapping for residual constraints on the electrode field.

$$H_{mod}(E) = \int_{\mathcal{M}} \nabla \cdot (\sigma(r) \nabla V) \, d\Omega \pmod{p_k}$$

Where $\mathcal{M}$ represents the structural cortical manifold, and $\sigma(r)$ denotes the conductivity tensor bounded by a prime modulus p_k . The application of Feynman Path Integrals maps sub-mW thermal dispersions.

$$K(x, y) = \lim_{N \rightarrow \infty} \sum_{paths} \exp\left(\frac{iS[x(t)]}{\hbar}\right) \equiv F_R(V_{stim})$$